

Hospitality Analytics Project

The Hospitality Analytics Project was undertaken to analyse performance, operational efficiency and revenue generation across a chain of luxury hotels operating in major Indian metro cities (Mumbai, Delhi, Bangalore and Hyderabad).

In response to declining market share and revenue leaks caused by intense strategic moves from competitors and inefficient direct-booking channels, this project aggregates booking transactional data across properties such as Atliq Grands, Atliq Exotica, Atliq City, Atliq Palace, Atliq Bay, etc. and multi-channel booking platforms such as Direct Online, Direct Offline, Journey, MakeMyTrip, LogTrip, Tripster, etc.

Key Highlights

- **Dataset Scope:** Realized bookings, cancellations, daily occupancy limits, capacity metrics, and revenue metrics across multiple room categories (Standard, Premium, Real Deluxe, Presidential).
- **Core Goal:** Provide actionable insights to optimize Average Daily Rate (ADR), Revenue per Available Room (RevPAR), Occupancy Percentage, and Direct Booking conversion while reducing cancellation-induced revenue leakage.

Project Objectives:

1. **Establish Standardized Hospitality Key Performance Indicators (KPIs):** Define and implement precise Dax and SQL logic for critical metrics such as
 - RevPAR
 - ADR
 - Occupancy %
 - DBRN (Daily Booked Room Nights)
 - DSRN (Daily Sellable Room Nights)
 - DURN (Daily Utilized Room Nights)
 - Realization %.

2. **Multi-Tool Analytical Architecture:**

This project implemented a Parallel Multi-Tool Analytics Framework, executing complete data cleaning, modeling, descriptive analytics, and dashboard deployment across four industry-standard analytics tools.

Comprehensive Process Standardization: The entire end to end analytics pipeline from initial data preparation and cleaning to model building and interactive dashboard

creation was performed across all four platforms to cross-validate insights and ensure data consistency.

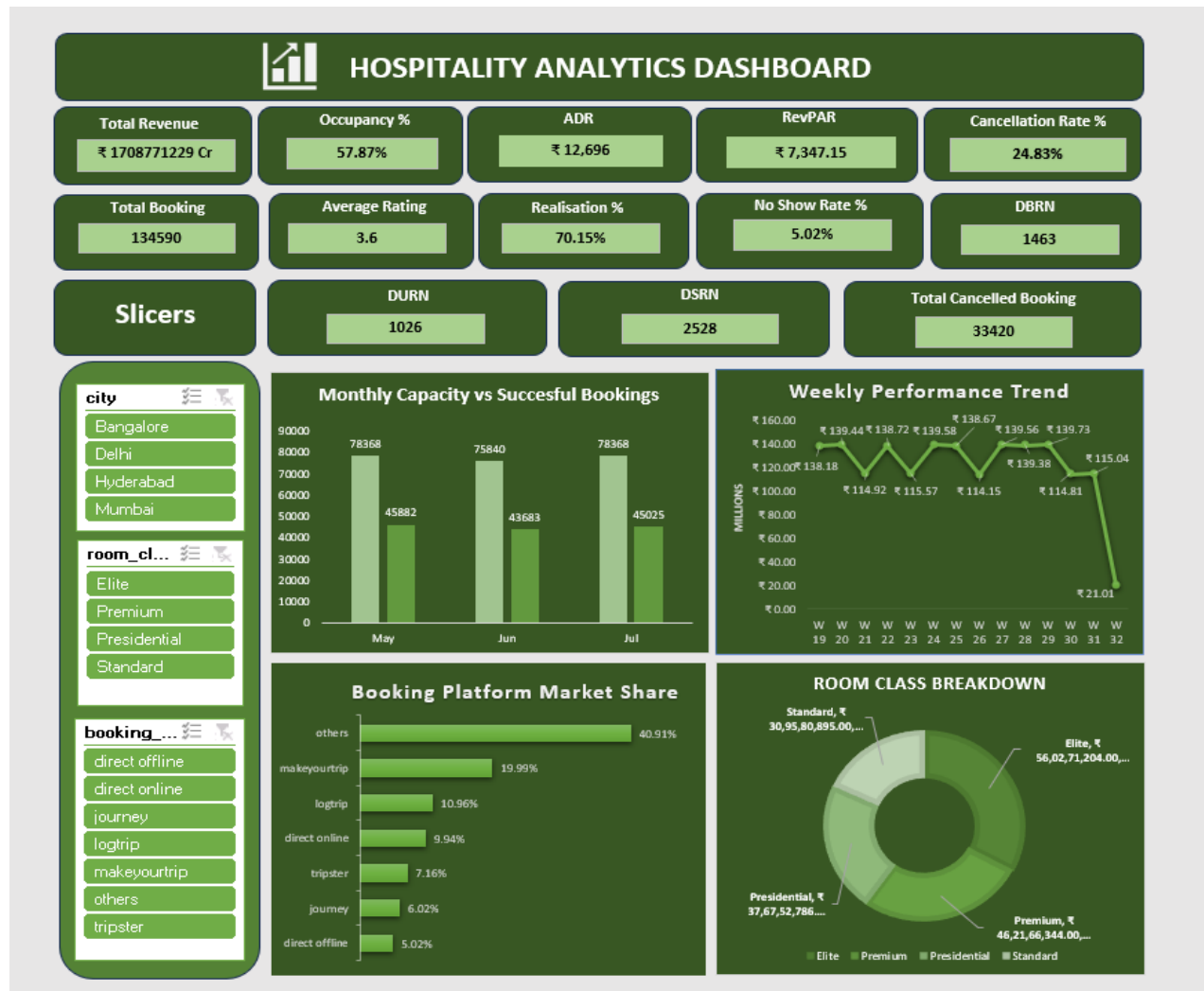
- **MySQL:** Leveraged for relational schema design, data cleaning queries, database normalization, window functions, and SQL-based KPI aggregations.
 - **Microsoft Excel:** Applied for exploratory data analysis (EDA), rapid data manipulation, formula-based data auditing, and preliminary pivot table structures.
 - **Power BI:** Utilized for data modeling, dynamic DAX (Data Analysis Expressions) calculations, dynamic slicers, interactive visuals, and cross-filtering dashboard layouts.
 - **Tableau:** Deployed for high-impact visual storytelling, spatial/geographical performance mapping, and specialized channel distribution dashboards.
3. **Identify Revenue Bottlenecks:** Diagnose root causes behind low realization rates, non-performing booking channels, and weekday/weekend pricing disparities.
4. **Strategic Action Plan and Recommendations:** Offer data driven recommendations to clients or stakeholders to improve property performance and pricing strategy of hospitality firm.

Major KPI Definitions & Simplified Business Equivalent:

Standard Metric	Industry Definition	Business Meaning
RevPAR	Revenue Per Available Room	Average earnings generated per total available room (booked or vacant).
ADR	Average Daily Rate	Average price actually paid per booked room per day.
Occupancy %	Total Booked / Total Capacity	Percentage of total available hotel capacity filled by guests.
Realization Rate (%)	Checked-Out Stays / Total Bookings	Percentage of bookings where guests actually checked in and stayed.
DBRN	Daily Booked Room Nights	Average number of room nights booked per day across properties.
DSRN	Daily Sellable Room Nights	Total daily available room supply capacity.
DURN	Daily Utilized Room Nights	Average number of room nights physically occupied by checked-out guests.

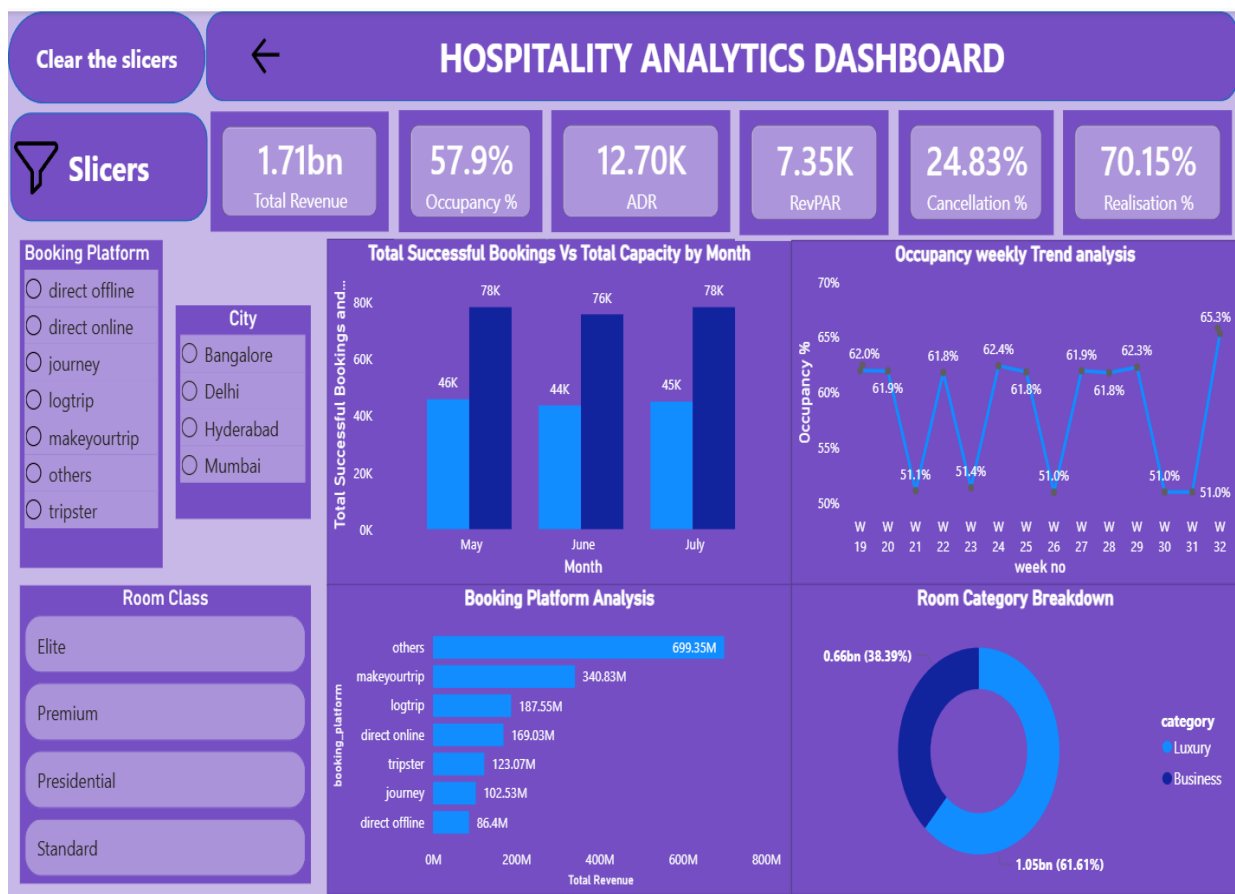
Dashboards created in various analytical tool:

1.Excel Dashboard:



- **Purpose:** Quick operational overview and dynamic pivot analysis.
- **Key Components:**
 - Top summary strip displaying Realized Revenue (₹ 170.88 Cr / ₹ 1.71 Bn), Occupancy Rate (57.87%), Average Daily Rate (₹ 12,696), RevPAR (₹ 7,347.15), and Cancellation Rate (24.83%).
 - Dynamic Pivot Tables enabling instant breakdown by property and room category (Standard, Premium, Real Deluxe, Presidential).
 - Interactive Slicers for City, Room Class and Booking platform.

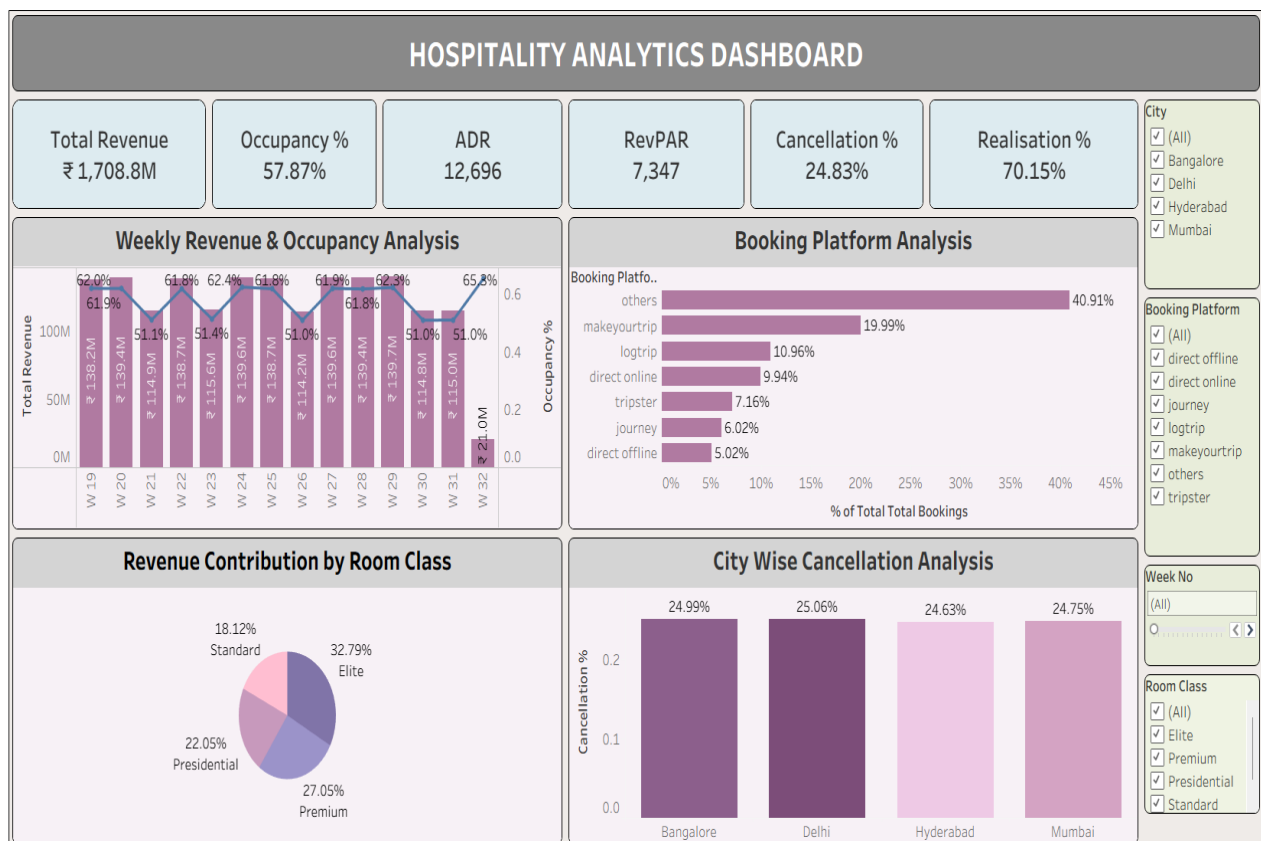
2.Power BI Dashboard:



- **Purpose:** Comprehensive multi-dimensional deep dive with DAX measures.
- **DAX Formulas Implemented:**
 - $\text{Total_Revenue} = \text{SUM}(\text{fact_bookings}[\text{revenue_realized}])$
 - $\text{Occupancy \%} = \text{DIVIDE}([\text{Total Successful Bookings}], [\text{Total Capacity}], 0)$
 - $\text{ADR} = \text{DIVIDE}([\text{Total Revenue}], [\text{Total Successful Bookings}], 0)$
 - $\text{RevPAR} = \text{DIVIDE}([\text{Total Revenue}], [\text{Total Capacity}], 0)$
 - $\text{Cancellation \%} = \text{DIVIDE}(\text{CALCULATE}(\text{COUNT}(\text{fact_bookings}[\text{booking_id}]), \text{fact_bookings}[\text{booking_status}] = "Cancelled"), [\text{Total Bookings}], 0)$
 - $\text{Realisation \%} = \text{DIVIDE}([\text{Total Checked Out}], [\text{Total Bookings}], 0)$
- **Key Visuals:**
 - **Total Successful Bookings Vs Total Capacity by Month:** A clustered column bar chart comparing actual successful bookings against available total room capacity across May, June, and July.

- **Occupancy Weekly Trend Analysis:** A line chart tracking fluctuations in Occupancy % over week numbers (Week 19 through Week 32).
- **Booking Platform Analysis:** A horizontal bar chart ranking revenue performance across platforms such as others, makeyourtrip, logtrip, direct online, tripster, journey, and direct offline.
- **Room Category Breakdown:** A donut chart showing revenue split between Business (1.05bn / 61.61%) and Luxury (0.66bn / 38.39%) categories.
- **Interactive Slicers & Filters:** Left-hand navigation elements allowing dynamic filtering by Booking Platform, City and Room Class

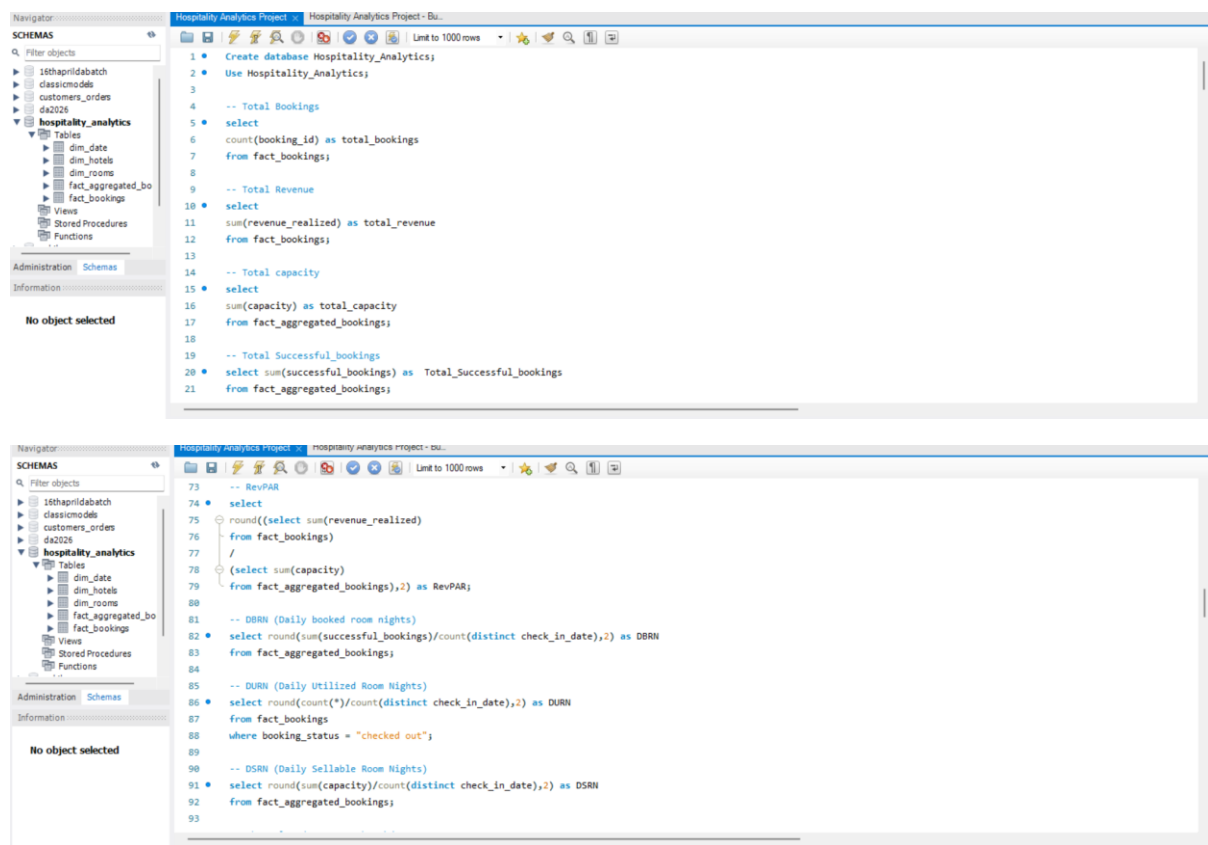
3.Tableau Dashboard:



- **Purpose:** Visual storytelling, spatial maps, and rating distributions.
- **Key Visuals:**
 - **Weekly Revenue & Occupancy Analysis:** A combo chart (bars for Total Revenue alongside a line plot for Occupancy %) tracking performance across Week 19 to Week 32.
 - **Booking Platform Analysis:** A horizontal bar chart illustrating the percentage distribution of total bookings across various booking channels (others at 40.91%, makeyourtrip at 19.99%, logtrip at 10.96%, direct online at 9.94%, tripster at 7.16%, journey at 6.02%, and direct offline at 5.02%).

- **Revenue Contribution by Room Class:** A pie chart detailing revenue percentage share across room categories: Elite (32.79%), Premium (27.05%), Presidential (22.05%), and Standard (18.12%).
- **City Wise Cancellation Analysis:** A column bar chart comparing cancellation rates across major markets—Bangalore (24.99%), Delhi (25.06%), Hyderabad (24.63%), and Mumbai (24.75%).
- **Interactive Slicers Panel:** A right-hand sidebar featuring filter options for City, Booking Platform, Week No, and Room Class.

4.SQL Analysis:



MySQL served as the relational data foundation for this project, enabling structured schema setup, data aggregation, and accurate metric validation before building downstream visual reports.

- **Relational Schema Architecture:** Established the core database structure (**Hospitality_Analytics**) utilizing a star-schema design with dimension tables (**dim_date**, **dim_hotels**, **dim_rooms**) and fact tables (**fact_bookings**, **fact_aggregated_bookings**).
- **Core Volume & Financial Aggregations:** Executed SQL queries to compute baseline operational totals, including Total Bookings (COUNT(booking_id)), Total Revenue (SUM(revenue_realized)), Total Capacity (SUM(capacity)), and Total Successful Bookings (SUM(successful_bookings)).

Advanced Hospitality KPI Calculations:

- **RevPAR (Revenue Per Available Room):** Formulated complex scalar subqueries to calculate total realized revenue relative to total available room capacity using `ROUND((SELECT SUM(revenue_realized) FROM fact_bookings) / (SELECT SUM(capacity) FROM fact_aggregated_bookings), 2)`
- **DBRN (Daily Booked Room Nights):** Calculated average daily successful bookings using `SUM(successful_bookings) / COUNT(DISTINCT check_in_date)`.
- **DURN (Daily Utilized Room Nights):** Measured actual room consumption by filtering completed stays via `WHERE booking_status = 'checked out'` against distinct operational dates.
- **DSRN (Daily Sellable Room Nights):** Determined daily inventory supply by dividing total room capacity by the distinct count of check-in dates.

Data Challenges & Resolution Strategies:

Challenge Encountered	Business Impact	Analytical Resolution Strategy
Data Inconsistency & Null Ratings	Unrated customer bookings skewed customer satisfaction (CSAT) metrics.	Cleaned null values; handled unrated records as a separate category (No Rating) to prevent baseline skewness.
Cancellation & No-Show Revenue Skew	Gross booking revenue overstated true business realization due to high cancellation rates (24-25%).	Created strict distinction between Total Revenue Generated (Gross) and Total Revenue Realized (Net of cancellations and no-shows).
Capacity vs. Utilization Variance	Raw booking counts did not account for property capacity ceiling differences.	Formulated calculated columns and DAX/SQL logic for Realized Capacity and Utilized Room Nights (DURN) vs. Sellable Room Nights (DSRN).
Dynamic Pricing Analysis Disparities	Pricing logic differed between weekend rate spikes and weekday baselines without explicit tags.	Engineered a custom date dimension table (dim_date) incorporating day_type (Weekend vs. Weekday) and week_no flags.

Core Business Insights Derived:

1. Mumbai Leads Revenue, Delhi Leads Occupancy:

- **Mumbai** generates the highest total realized revenue (~40% of chain total), driven by high ADR in premium properties (Atliq Palace and Atliq City).
- **Delhi** exhibits the highest average occupancy rate (~60-62%), but has a lower ADR compared to Mumbai.

2. High Cancellation Leakage (24.8%):

- Nearly 1 in every 4 bookings gets cancelled.
- Online Travel Agencies (OTAs) like MakeMyTrip and LogTrip exhibit significantly higher cancellation rates (27%) compared to Direct Corporate or Direct Online bookings (18-20%).

3. Underutilized Weekend Pricing Opportunity:

- Weekend occupancy surges by 15–20% compared to weekdays, yet ADR remains mostly static across properties. This indicates **static pricing practices**, leading to missed revenue opportunities on peak days.

4. Direct Channel Underperformance:

- Direct booking channels (Direct Online and Direct Offline) contribute less than 15% to total revenue, indicating heavy dependence on third-party OTAs, which incurs commission fees of 15-22%.

5. Low Ratings Impact on Occupancy:

- Properties with average customer ratings below 3.0 (e.g., Atliq Season) experience a ~18% lower occupancy rate and significantly higher cancellation rates.

6. OTA Commission Friction:

- Third-party platforms (like MakeMyTrip and LogTrip) control ~31% of your bookings but carry higher cancellation rates (~27%). You are paying high commission fees (15–22%) on revenue that has a high chance of vanishing before check-in.

7. City-Level Cancellation Consistency:

- Cancellation rates across all four major cities are nearly identical (~24.6% to 25.0%). This indicates that cancellations are driven by overall platform policy or customer behaviour rather than location-specific issues.

Strategic Recommendations for Clients:

Based on the findings from SQL queries, Power BI models, and visual analyses:

1. Implement Dynamic Pricing Mechanics

- **Action:** Shift from fixed rates to dynamic pricing algorithms.
- **Impact:** Increase weekend ADR by **10–15%** in high-demand cities (Mumbai and Bangalore) where weekend occupancy consistently exceeds 70%.

2. Revamp Cancellation & No-Show Policies

- **Action:** Introduce tiered non-refundable booking rates for third-party OTAs and charge a nominal fee for late cancellations (within 24–48 hours of check-in).
- **Impact:** Reduce cancellation rate from **24.8% to under 18%**, reclaiming substantial lost revenue.

3. Direct Channel Conversion Program

- **Action:** Offer direct booking incentives such as free breakfast, room upgrades, or loyalty points on Atliq Direct website.
- **Impact:** Increase direct booking share from **15% to 28%**, saving up to 5-8% in OTA commission costs.

4. Focused Quality Overhaul for Low-Rated Properties

- **Action:** Audit operational and service bottlenecks at low-rated properties (ratings < 3.0) to address customer complaints regarding room service and cleanliness.
- **Impact:** Elevating ratings to 4.0+ will restore occupancy levels by **12–15%**.

5. Implement "Book Direct & Save" Loyalty Perks:

- **Action:** Instead of competing purely on price against OTAs, offer direct bookers non-monetary perks such as complimentary breakfast, room upgrades, flexible check-outs, or loyalty points.
- **Impact:** Increase direct booking share from **15% to 28%**, saving 5–8% in net commission expense while building direct customer relationships.

6.Strategic Overbooking Protocol for High-Cancellation OTA Channels:

- **Action:** Because OTA slots experience a high (~27%) cancellation rate, implement a controlled 5–8% overbooking threshold specifically for non-prepaid OTA slots during high-demand weekdays.

- Expected Impact: Maximizes physical room occupancy and buffers against last-minute channel drop-offs.

Conclusion:

This **Hospitality Analytics Project** demonstrates how combining multi-tool data engineering (MySQL), interactive modeling (Power BI), rapid analysis (Excel), and visual analytics (Tableau) provides end-to-end visibility into hotel performance. Implementing these recommendations will help the client optimize yield, protect revenue against cancellations, and strengthen direct sales channels.